137 MHz V-dipole for weather satellites

Two quarter-wave rods at 120° in a horizontal plane, fed by an SMA jack in a 3D-printed hub on a PVC mast: the 9A4QV receive antenna for NOAA APT and Meteor LRPT near 137 MHz.

✓ All 16 checks pass

Make it yours

Centre frequency 137.5 MHz · V angle 120 ° · Element diameter 6 mm solid rod · Element metal Aluminium rod (6060 / 6063), tapped and screwed · Mast pipe PVC-U pressure pipe, 40 mm outside, PN16 · Design wind gust 35 m/s gusts (exposed roof or garden) · Mast pipe length 1200 mm · Coax length 3000 mm · Choke Five turns of the coax round the mast

OVERALL
940 × 340 × 1221 mm
length × depth × height

MASS
1.0 kg
as built, with every part listed

PIECES
3 cut (2 different)

WIND ON MAST, HUB AND ELEMENTS
factor 2.7
safety factor on yield

COAX SNAGGED AT THE HUB
factor 3.2
safety factor on yield

3D PRINTS
2 parts
about 66 g of filament, 5.5 h printing

BUILD TIME
7 h
7 steps

COST
~$51
everything on the materials list

FREQUENCY
137.5 MHz
centre; NOAA APT and Meteor LRPT

ELEMENT
534 mm
each, aluminium ø6

V ANGLE
120°
opening north or south

TIP TO TIP
937 mm

HUB HEIGHT
1.22 m
above the bottom of the mast

Contents

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1 Checks

All 16 checks pass at this size.

	Check	Result
✓	**Every print fits the printer** As it sits on the bed, turned the way the files come.	Feed hub: 80 × 93 × 68 on a 220 × 220 × 250 bed
✓	**Nothing floating** Every member, plate and part touches what holds it.	all connected
✓	**No clashes** No member passes through another away from a joint, and no plate through a member it is not notched around.	none
✓	**Every piece fits the stock** Cut from standard lengths with a trim and a kerf per cut.	longest 1200 mm
✓	**Every piece can be clamped and cut** No piece shorter than 32 mm on its short side.	shortest 534 mm
✓	**Wind on mast, hub and elements: strength** Yield ÷ peak stress with 45 N of drag in 35 m/s gusts (34 N on the mast, 11 N on hub and rods, at the top), plus self weight.	safety factor 2.67 (need 2), worst B (mast)
✓	**Coax snagged at the hub: strength** Yield ÷ peak stress with 25 N pulled sideways at the hub, plus self weight.	safety factor 3.15 (need 2), worst B (mast)
✓	**Every part has a step** Each piece is cut, fitted or bought in a numbered step.	7 steps
✓	**Element length from frequency** A rod resonates a little shorter than a free-space quarter wave; the end effect for a 6 mm rod is about 2.0 %. Tuning in step 6 trims the rest.	λ/4 at 137.5 MHz = 545 mm; × 0.980 = 534 mm per element
✓	**V angle** At 120° the horizontal V has its most gain overhead and a usable lobe down to low passes, and its impedance comes close to 50 Ω. Narrower drops the impedance; 180° is a plain dipole with nulls off its ends.	120° between the elements
✓	**Feed gap** Small against the wavelength so it does not detune the feed, and wide enough that the two feed joints cannot touch (at least 4 mm).	12 mm between rod ends (0.55 % of a wavelength), 6 mm between the screw heads
✓	**Feed match (rough estimate)** Estimate R ≈ 73 Ω × sin²(angle ÷ 2) at resonance; the NanoVNA in step 6 gives the real number. Under 2 is plenty for receiving.	about 55 Ω, SWR 1.09 on 50 Ω coax
✓	**Choke on the coax** At least 500 Ω of common-mode impedance keeps current off the outside of the shield, so the coax does not become part of the antenna and pick up house noise. Coil from Wheeler's formula; ferrites from the datasheet curve.	5 turns, about 1.10 µH, 951 Ω at 137.5 MHz
✓	**Coax loss** Under 3 dB keeps the weak satellite signal above the receiver noise; use a better cable (or a preamp at the hub) for a longer run.	0.8 dB over 3.00 m of RG58 with its plugs
✓	**Elements in the wind** Each rod as a cantilever 499 mm out of its arm in 35 m/s gusts plus its own weight, against the yield of aluminium.	safety factor 5.30 (need 2); tip sags 1.3 mm under its own weight
✓	**Printed hub fits** Push fits: bore 0.4 mm over the size; at least 2.5 mm of wall round a bore that is clamped; fits a 200 mm bed.	socket ø40.4 mm on a ø40 mm pipe, rod bores ø6.4 mm on ø6 mm rod, 3.8

		mm wall round each rod; 80 × 93 on the bed

2 Materials

Stock

Section	Pieces	Cut length	Buy	Mass	Cost
ø6 solid aluminium rod	2	1.1 m	2 × 1.00 m	0.1 kg	$6.00
40 × 3 PVC-U pressure pipe, PN16	1	1.2 m	1 × 3.00 m	0.6 kg	$14

3D prints

Part	Settings	Qty	Filament	Print	Step	Cost	Files
Feed hub	PETG · 0.2 mm layers · 4 walls · 40% infill · no supports	1	61 g	5.1 h	1	$1.35	
Pocket lid	PETG · 0.2 mm layers · 4 walls · 40% infill · no supports	1	4 g	0.4 h	1	$0.09	

Filament and time are estimates from the part's volume; your slicer has the real numbers. The .stl is the part at this size, in millimetres; the .scad is its OpenSCAD source.

Parts to buy

Part	Spec	Qty	Step	Cost
Heat-set insert, M3	M3, short (about 4 long), for a 4.0 hole	4	1	$0.40
SMA female bulkhead jack, solder cup	panel mount, 50 Ω, with nut and lock washer, for panels up to 4 thick	1	2	$2.00
Solder tag to fit the SMA jack thread	tinned brass ground lug	1	2	$0.20
Ring terminal, M3	tinned copper, solder or crimp	2	2	$0.20
Hook-up wire	stranded, silicone insulated, 0.5 square	1	2	$0.50
Screw, M3 × 6, pan head, stainless	with a stainless washer, into the tapped rod ends	2	3	$0.20
Grub screw, M3 × 6, cup point, stainless	into the heat-set inserts, onto the rods	2	3	$0.20
Bolt, M4 × 60, stainless, with nyloc nut and two washers	through the hub socket and the mast	1	4	$0.40
Coax, RG58, SMA male to SMA male	3.0 metre assembly (or the plug your receiver needs at the far end)	1	5	$9.60
Cable ties, UV-stable black	200 long	10	5	$0.50
Screw, M3 × 8, pan head, stainless	lid into the heat-set inserts	2	7	$0.20
Mast clamp (U-bolt with saddle)	M6 U-bolt for a 40 pipe, with a saddle for your post or wall bracket	2	7	$6.00

Consumables

- Solder, rosin core · a little
- Self-amalgamating tape · 1 roll
- PVC electrical tape, black · 1 roll
- Neutral-cure silicone sealant · 1 small tube
- Cutting oil for tapping aluminium · a few drops

Tools

- 3D printer and the filament named with each print
- Soldering iron (with a heat-set insert tip if you have one)
- Hacksaw or fine-tooth saw, file and deburring tool
- Drill, centre punch, the two bits named in steps 3 and 4, M3 tap and tap wrench
- Calipers and tape measure

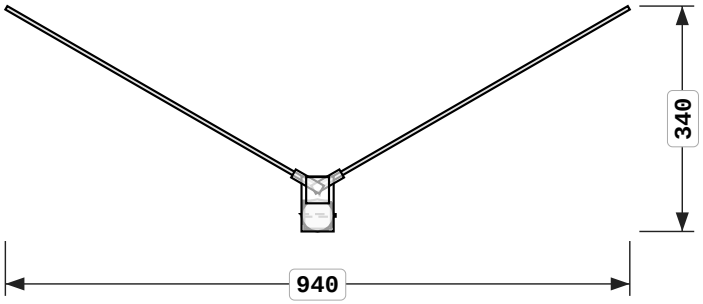
- Multimeter
- NanoVNA or antenna analyser
- Spanner for the SMA nut, screwdrivers, hex key for the grub screws
- Spirit level and compass

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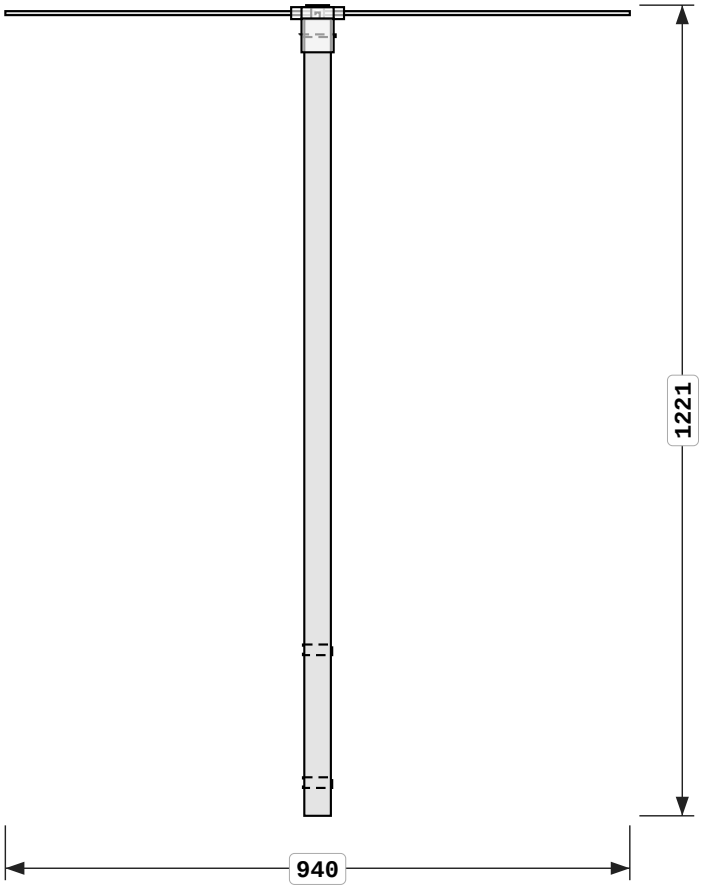
3 Drawings

Overall dimensions. Every number on this page is from the same model as the 3D view.

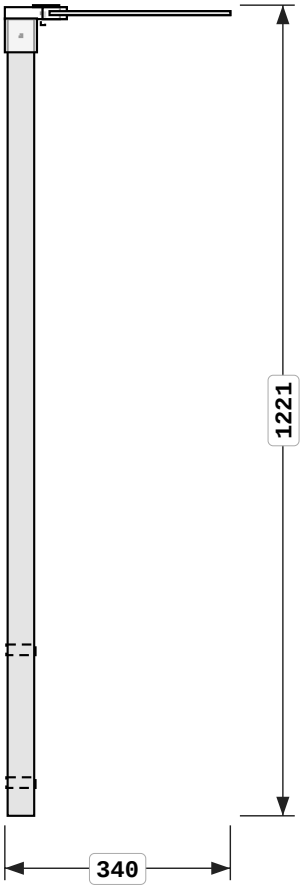
Top



Front



Side (from the right)



4 Legend

Units	millimetres, kilograms, degrees
Datum	x east from the apex of the V (the feed point), y north (the V opens toward north), z up from the bottom end of the mast
Piece marks	A, B, C… one letter per different piece; x4 after it means cut four the same. Mirror images share a mark.
Cut marks	From one square end, mark each edge at the two numbers, join the marks around the section, cut on the lines. Angles are for reference.
Edges	named by where they face on the finished build (top-south, bottom-east…)
Joints	nothing is welded: parts clamp, as each step says
Long point	the overall length of the piece, tip to tip
Dimensions	jig drawings and joint tables give centre lines (℄), where the members' centres cross; the general arrangement gives outside sizes
Tolerance	cut ±1 mm unless a step says otherwise

5 Order

Start the print, then cut and tap the rods and the mast while it runs; wire the jack, fit the rods, tune, then seal and mount.

Fabricate

1. Print the hub and the lid 1 h
2. Wire the SMA jack 0.5 h
3. Elements: cut, tap and fit 1.5 h
4. Mast and hub 1 h

Assemble

5. Coax and coil choke 0.5 h

Tune and install

6. Tune with a NanoVNA 1 h
7. Seal, mount and point it 1.5 h

6 Steps

Ticks save on this device. "Show in 3D" shows the build after that step, the new parts lit.

01 Print the hub and the lid

Fabricate · 1 h

- [ ] Print the feed hub and the pocket lid from the files below (settings and orientation with each). PETG or ASA, not PLA: it lives outdoors.
- [ ] Start the print first: it takes a few hours, and steps 2 to 4 can be done while it runs.
- [ ] Clean the strings out of the rod bores, the SMA hole and the socket. Run a drill of the bore size through each rod bore by hand if a rod does not slide in.
- [ ] Melt an M3 heat-set insert into each of the four 4 mm holes on the top face with a soldering iron, flush with the face: two over the arms (grub screws), two behind the pocket (lid).

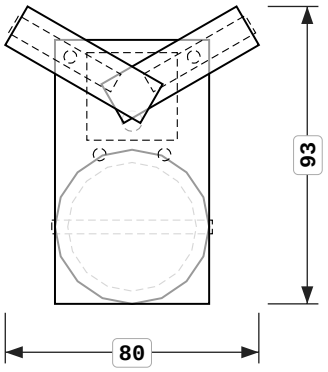
Check A rod slides into each arm bore by hand without play you can feel; the mast pipe pushes into the socket to the end; all four inserts square and flush within ±0.5 mm.

> A bore that came out tight is drilled out, never forced: forcing a rod in splits the arm along its layers.

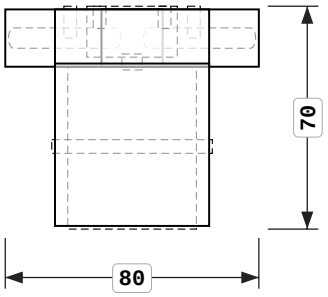
Print: Feed hub

PETG · 0.2 mm layers · 4 walls · 40% infill · no supports · about 61 g, 5.1 h each · print it top face down on the bed, socket pointing up (the files come turned this way); no supports, the pocket floor bridges

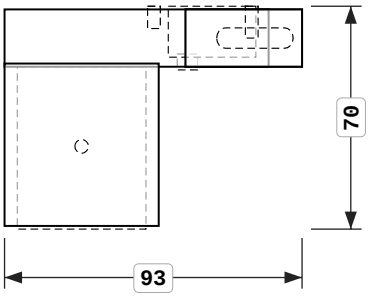
Feed hub: Top



Feed hub: Front



Feed hub: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø40.4	z	39.8, 24.2, -1	51
Hole	ø4.3	x	14.6, 24.2, 25	50.4
Pocket	28.4 × 27.5 × 16		39.8, 65.1, 61	
Hole	ø6.4	z	39.8, 57.4, 49	5
Hole	ø6.4	angled	44.9, 69.4, 59	37
Hole	ø6.4	angled	34.7, 69.4, 59	37
Hole	ø4	z	59.2, 77.6, 59	10
Hole	ø4	z	20.4, 77.6, 59	10
Hole	ø4	z	29.6, 46.9, 62	7
Hole	ø4	z	50, 46.9, 62	7

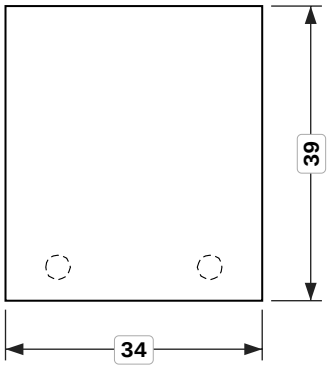
Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn; the files come turned for printing (+z face on the bed). On the bed it is 79.6 × 93.4 × 68 mm.

> Melt an M3 heat-set insert into each of the four small holes on the top face.

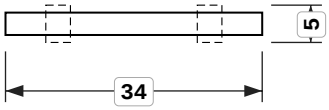
Print: Pocket lid

PETG · 0.2 mm layers · 4 walls · 40% infill · no supports · about 4 g, 0.4 h each · print it flat on the bed

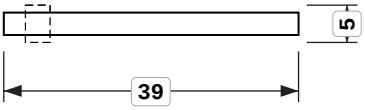
Pocket lid: Top



Pocket lid: Front



Pocket lid: Side (from the right)



Cut	Size	Axis	From (x, y, z)	Depth
Hole	ø3.3	z	7, 4.5, -1	5
Hole	ø3.3	z	27.4, 4.5, -1	5

Dashed on the drawing. Positions from the part's lower-left-bottom corner as drawn. On the bed it is 34.4 × 39.5 × 3 mm.

> Screwed down over a bead of neutral-cure silicone after tuning.

Parts

- Heat-set insert, M3 ×4 M3, short (about 4 long), for a 4.0 hole

02 Wire the SMA jack

Fabricate · 0.5 h

- [ ] Slip the solder tag over the thread of the SMA jack, then push the jack down through the hole in the floor of the pocket from inside, thread first, so the jack points down under the hub.
- [ ] Fit the lock washer and nut from underneath and tighten with a spanner while holding the jack body inside: it must not turn.
- [ ] Cut two tails of hook-up wire 40 mm long, strip 4 mm at each end, and solder an M3 ring terminal on one end of each.
- [ ] Solder one tail to the centre solder cup of the jack (this is the east element feed) and the other to the solder tag (shield, west element).
- [ ] Bend the tails so each free end lies over where its rod end will be: the centre tail to the east (right, looking north), the tag tail to the west.

Check Multimeter: centre pin to jack body open circuit (no short); centre pin to the east tail and jack body to the west tail both under 1 Ω.

> Heat the solder cup briefly: the jack's plastic insulator melts if you hold the iron on for long.

Parts

- SMA female bulkhead jack, solder cup x1 panel mount, 50 Ω, with nut and lock washer, for panels up to 4 thick
- Solder tag to fit the SMA jack thread x1 tinned brass ground lug
- Ring terminal, M3 ×2 tinned copper, solder or crimp
- Hook-up wire x1 stranded, silicone insulated, 0.5 square

03 Elements: cut, tap and fit

Fabricate · 1.5 h

- [ ] Cut the two elements (A) from 6 mm aluminium rod, square ends, to the length on the cut sheet. Deburr both ends of each.
- [ ] In one end of each rod, centre-punch, drill 2.5 mm on the axis 8 mm deep, and tap M3 with a drop of cutting oil. This is the inner (feed) end.
- [ ] Push each rod, feed end first, into an arm from outside until the feed end stands in the pocket.
- [ ] Screw an M3 × 6 screw with a washer and its ring terminal into each rod end: centre tail to the east rod, tag tail to the west rod. Snug, not tight.
- [ ] Slide the rods until the end faces are 12 mm apart (calipers between them, across the pocket), then tighten a grub screw down onto each rod through the insert over its arm.

Check Both rods the same length within ±1 mm; end faces 12 ±1 mm apart; tip to tip 937 ±5 mm; each rod tight when you twist it.

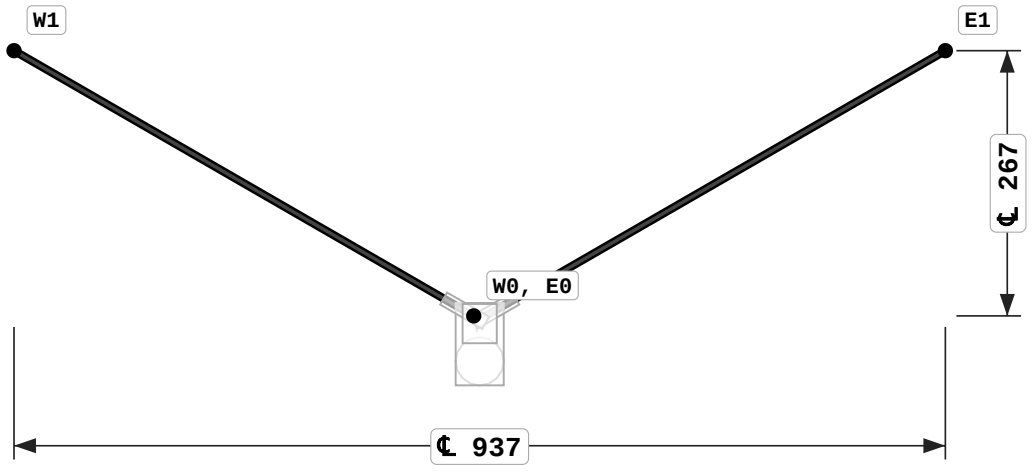
> The element length is measured from the rod's inner end face. Do not cut them shorter "to be safe": tuning in step 6 trims them, about 3.9 mm per MHz.

Parts

- Screw, M3 × 6, pan head, stainless x2 with a stainless washer, into the tapped rod ends
- Grub screw, M3 × 6, cup point, stainless x2 into the heat-set inserts, onto the rods

Jig

Step 3: Top



Joint positions

Point	x	y	z
W0	-6	3	1209
E0	6	3	1209
W1	-469	271	1209
E1	469	271	1209

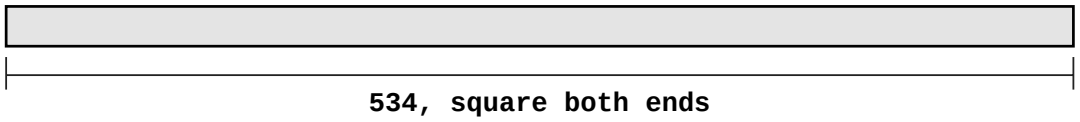
Pieces: A x2

A x2 — **534 mm**

Element · Aluminium rod ⌀6 · step 3

long point

`elem-E, elem-W`



Left end: free end · Right end: free end

04 Mast and hub

Fabricate · 1 h

- [ ] Cut the mast (B) to length from the 40 mm PVC pipe with a fine saw, square; deburr inside and out.
- [ ] Mark a line 25 mm down from the top end and drill a 4.3 mm hole straight through the pipe on it, across a diameter.
- [ ] Push the hub onto the top of the mast until the pipe bottoms in the socket, turn it until the cross holes line up, and fit the M4 bolt with a washer each side and the nyloc nut.
- [ ] Mark the clamp positions on the mast: 50 and 250 mm up from the bottom end.

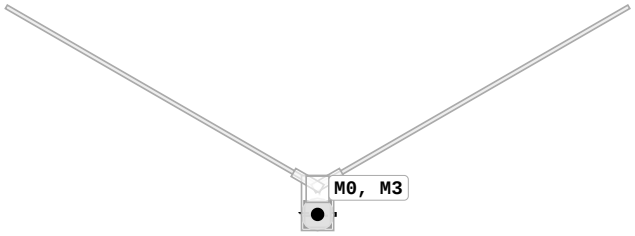
> **Check** Mast within ±1 mm of its length; the hub does not turn or lift on the mast by hand; the elements are level when the mast is held plumb (a spirit level on each rod).

Parts

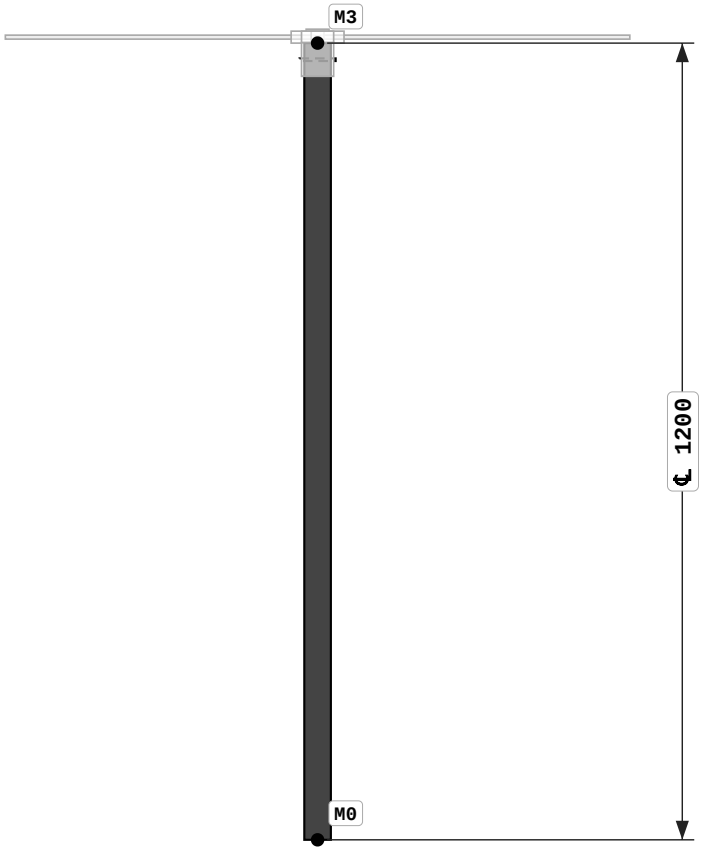
- Bolt, M4 × 60, stainless, with nyloc nut and two washers x1 through the hub socket and the mast

Jig

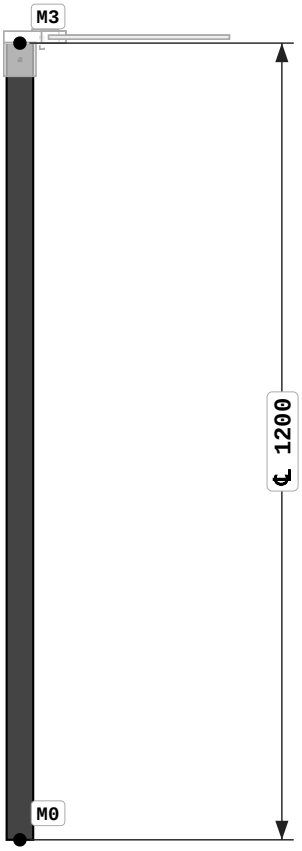
Step 4: Top



Step 4: Front



Step 4: Side (from the right)



Joint positions

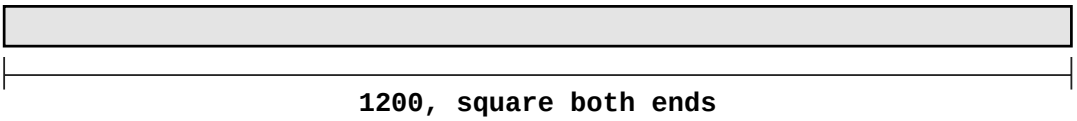
Point	x	y	z
M0	0	-42	0
M3	0	-42	1200

Pieces: B x1

B x1 — **1200 mm**

Mast · PVC 40×3 · step 4 — long point

mast



Left end: free end · Right end: free end

05 Coax and coil choke

Assemble · 0.5 h

- [ ] Screw the SMA plug of the coax onto the jack under the hub, finger tight, then an eighth of a turn with a spanner.
- [ ] Wrap the SMA joint in self-amalgamating tape, stretched, half lapped, from the hub underside down onto the cable; then a layer of PVC tape over it against UV.
- [ ] Directly under the hub socket, wind 5 tight turns of the coax round the mast, side by side (about 25 mm of mast), and hold the coil with tape at each end.
- [ ] Run the coax down the mast and cable-tie it every 200 mm; leave a drip loop where it leaves the mast.

Check 5 turns, tight against each other and against the mast, starting within 50 mm of the hub; nothing slides when you tug the cable.

Parts

- Coax, RG58, SMA male to SMA male x1 3.0 metre assembly (or the plug your receiver needs at the far end)
- Cable ties, UV-stable black x10 200 long

06 Tune with a NanoVNA

Tune and install · 1 h

- [ ] Clamp the mast upright in the open, elements at least 1.5 times their own length from walls, metal and people (a temporary post, a fence, a tripod).
- [ ] Calibrate the NanoVNA (open, short, load) at the end of its own short lead, then connect the far end of the antenna coax.
- [ ] Sweep 120 to 160 MHz and read the frequency of the lowest SWR. The aim is 137.5 MHz ±1 MHz.
- [ ] If the dip is low (below 136.5 MHz), the elements are long: file the outer tip of each rod back by the same amount, about 3.9 mm per MHz, deburr, and sweep again.
- [ ] If the dip is high, the elements are short: leave them if SWR at 137.1 and 137.9 MHz is under 2; otherwise cut a new pair 3.9 mm longer for each MHz it is high.
- [ ] No NanoVNA? An SWR meter needs a transmitter, and you may not transmit at 137 MHz. A licensed operator can read SWR at 144 and 146 MHz instead: it should be low at 144 and rise toward 146.

Check SWR minimum within ±1 MHz of 137.5 MHz and SWR under 2 from 137.1 to 137.9 MHz; both elements still the same length within ±1 mm.

> Trim both elements equally and a little at a time: a millimetre is easy to take off and impossible to put back.

07 Seal, mount and point it

Tune and install · 1.5 h

- [ ] Fill the pocket with neutral-cure silicone (not acetic: it corrodes the joints) over the screws and the jack, lay the lid on, and screw it down with the two M3 × 8 screws.
- [ ] Fix the mast to its post, rail or wall bracket with the two U-bolt clamps on the marks, 50 and 250 mm up from the bottom end. Both clamps tight: the wind check counts on them.
- [ ] Turn the mast so the V opens toward north or south (true north: correct a compass for the local declination). East-west passes are what the open side faces away from.
- [ ] Set the mast plumb so both elements are level, and put it where it sees as much of the sky as you can, above the roof line if possible and away from other antennas.
- [ ] Run the coax to the receiver with the drip loop below the entry point.

Check Opening within 10° of north or south; each element level within 3° (a spirit level bubble inside the lines); the mast does not move when you shake it at the hub.

> Mount it lower rather than in a place where the mast cannot be clamped at both marks: the wind check assumes two tight clamps on a post that does not move.

Parts

- Screw, M3 × 8, pan head, stainless x2 lid into the heat-set inserts
- Mast clamp (U-bolt with saddle) x2 M6 U-bolt for a 40 pipe, with a saddle for your post or wall bracket

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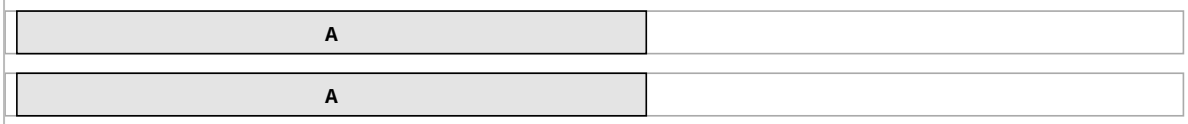
7 Cut list

Mark	Qty	Section	Long point	Left end	Right end	Step
A Element	2	Aluminium rod ø6	534	square · free end	square · free end	3
B Mast	1	PVC 40×3	1200	square · free end	square · free end	4

Cutting plan

Each bar is one stock length; cut the pieces left to right.

Aluminium rod ⌀6 · 2 × 1.00 m lengths · kerf 2



PVC 40×3 · 1 × 3.00 m lengths · kerf 2



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137 MHz V-dipole for weather satellites · v1. Generated by the blueprint skill. Check every dimension against your own stock before you cut, wear eye and hand protection, and load-test the build gently before trusting it.